

nostr-veil — Submission

The problem in plain English

Every reputation system on the internet has the same flaw: you can see who endorsed whom.

That sounds harmless for restaurant reviews or follower counts. It becomes dangerous the moment the subject is sensitive. A whistleblower's colleagues know they're being watched. An abuse victim's friends are afraid to vouch for them publicly. A political dissident cannot risk being listed as a trusted contact.

The people who need reputation systems most are the people who cannot afford to use them.

What is a Web of Trust?

A Web of Trust (WoT) is how decentralised networks decide who to trust when there is no central authority to ask.

Instead of a company like Twitter deciding your credibility score, trust flows from people you already trust. If Alice trusts Bob, and Bob vouches for Carol, Alice can extend a degree of trust to Carol. Each node in the graph is a person; each edge is a trust assertion.

Nostr is a decentralised social protocol used by hundreds of thousands of people. It has a standard for publishing these trust scores called **NIP-85 Trusted Assertions**: kind `30382` events where a trusted provider publishes scores (rank, follower count, engagement) for any pubkey.

The problem with NIP-85 today: every assertion is signed with the publisher's key. The trust graph is fully public. Anyone can see who is vouching for whom.

What nostr-veil does

nostr-veil adds one cryptographic layer to NIP-85: **ring-signature privacy**.

A group of trusted people form a **trust circle**. Each member scores a subject independently. Their individual scores are wrapped in a **Linkable Spontaneous Anonymous Group (LSAG) ring signature** — a cryptographic construction that proves *someone in the circle* signed, without revealing *which* member.

The aggregated result is published as a completely standard NIP-85 `kind 30382` event. Any existing Nostr client reads it unchanged. Veil-aware clients can additionally verify the cryptographic proofs embedded in

the `veil-ring` , `veil-threshold` , and `veil-sig` tags.

Before: Alice signs → "I endorse Bob, rank 85" (Alice is identified)

After: Circle signs → "3 of 8 circle members endorse Bob, rank 80"

Verifiable: yes Signer identity: unknown

Key properties:

Property	NIP-85 today	nostr-veil
Verifiable by anyone	✓	✓
Backwards compatible with existing clients	✓	✓
Contributor identity hidden	×	✓
Double-signing prevented	×	✓

The LSAG linkability property means the same person cannot contribute twice to the same circle: each signature produces a unique **key image** that is checked on verification. You can prove someone was in the ring; you cannot prove which member they were.

The interactive demo

Live demo: <https://nostr-veil.pages.dev>

The demo tells a concrete story: a group of journalists anonymously scoring a whistleblower source's credibility before publishing. The scenario is chosen deliberately — it is the use case where deanonymisation causes the most harm.

What you will see

Step 1 — The Circle. Eight journalists form a trust circle. The user takes one journalist's role. The circle ID is derived deterministically from the sorted member pubkeys: change the membership and the ID changes.

Step 2 — The Source. An anonymous source has come forward claiming to have leaked documents from a major corporation. The journalist assigns a credibility score (0–100).

Step 3 — The Veil. Five of the eight journalists contribute anonymous LSAG-signed assessments. A `kind 30382` NIP-85 event is constructed live, showing the `veil-ring` , `veil-threshold` , and `veil-sig` tags being assembled in real time. The median score is aggregated. No individual score is visible.

Step 4 — Verification. The ring signature proof is verified step by step: extract the ring, parse each LSAG signature, verify against the ring, check key images for double-signers, confirm threshold. Result: **VALID — 5 distinct signers in a circle of 8**.

Step 5 — The Reveal (optional). One journalist wants to claim credit for being in the circle — perhaps for a journalism award, or legal protection. Using **nsec-tree** (a companion library for deterministic sub-identity derivation), the journalist proves they own both their anonymous circle persona and their public identity, without revealing which score was theirs. The other seven journalists remain anonymous.

Step 6 — The Network. A live trust graph shows all assertions, ring endorsements, and canary heartbeats. Two interactive demonstrations:

- **Ring Endorse** — shows an anonymous NIP-85 assertion flowing through the ring as an animated LSAG arc
- **Duress** — simulates a coerced member. Their heartbeat (canary-kit) stops. The network isolates the node, severs edges, and warns adjacent members. Ring signatures alone are not sufficient; coercion resistance requires liveness too.

How the cryptography works (non-technical)

Think of a ring signature like a signed note that says "one of these ten people wrote this" — but the note itself gives no clue which one. The mathematics ensure you cannot work backwards.

LSAG adds one property on top: **linkability**. Every signature produces a unique "key image" that is deterministic per private key. If the same person signs twice, their two key images will match. A verifier can detect double-voting without knowing who voted.

The signed message includes the circle ID and the subject pubkey, which means signatures from one circle cannot be replayed in another.

The broader ecosystem

nostr-veil is a thin integration layer. The real work is in the primitives beneath it:

Library	What it does	Independently useful
<code>@forgesworn/ring-sig</code>	SAG / LSAG ring signatures on secp256k1	✓
<code>nsec-tree</code>	Deterministic sub-identity derivation from a master key	✓
<code>canary-kit</code>	Coercion-resistant liveness and duress detection	✓
<code>signet</code>	Decentralised identity verification for Nostr	✓
<code>dominion</code>	Epoch-based encrypted access control	✓

Each library is maintained independently and is generally useful outside of nostr-veil. The maintenance surface for nostr-veil itself is small: if the underlying `@forgesworn/ring-sig` primitive is maintained, nostr-veil stays current at low cost.

NLnet and OpenSats funding applications are pending for `@forgesworn/ring-sig`, `nsec-tree`, and `signet`.

Who needs this

- **Journalists** verifying source credibility without exposing their network
- **Abuse reporting platforms** where reporters fear retaliation
- **Whistleblower networks** where association is itself dangerous
- **Political dissidents** in environments where trust relationships are surveilled
- **Anonymous peer review** where pre-publication bias is a concern

The pattern generalises: any context where the score itself is useful but the scorer's identity creates risk.

Technical summary

- **Language:** TypeScript, ESM-only
- **Cryptographic primitive:** LSAG on secp256k1 via `@forgesworn/ring-sig`
- **Nostr protocol:** NIP-85 (`kind 30382–30385`, `kind 10040`)
- **Output format:** standard Nostr event, fully backwards compatible
- **Tests:** 118 tests across 11 files

- **Package:** `nostr-veil` on npm
- **Source:** github.com/forgesworn/nostr-veil
- **Demo:** nostr-veil.pages.dev